

OJAS J. KIDILAY

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Summary

Mechanical Engineering graduate (B.Tech, 2026) specializing in computational simulation, physics-informed machine learning, and CAE-driven design validation. Co-author of a peer-reviewed paper accepted at ASSET 2026 on Physics-Informed Neural Networks (PINNs) for thermal analysis. Proficient in Ansys Fluent and OpenFOAM for CFD, with hands-on experience in turbulence modeling, mesh-independence validation, and benchmarking simulation results against published and theoretical data. Strong foundation in heat transfer, fluid mechanics, and structural analysis, combined with CAD expertise in SolidWorks, CATIA, and Creo. Smart India Hackathon 2023 Runner-Up.

Education

Bajaj Institute of Technology , Wardha Jul 2022 – Jun 2026
Bachelor of Technology in Mechanical Engineering Maharashtra, India

Relevant Coursework

• Strength Of Materials • Machine Design • Theory of Machines • Finite Element Analysis • Fluid Mechanics

Experience

Bajaj Engineering Skills Training Nov 2025 – May 2026
Graduate Trainee Engineer

- Designed a hydraulic steering circuit for a heavy-vehicle mechanism, sizing pump, control valve, and actuator for 150 bar / 12 L min⁻¹ operating conditions, and validated pressure-drop analysis across three load cases in SolidWorks, achieving simulated actuator response within $\pm 5\%$ of theoretical values.
- Developed PLC ladder logic and a 3-screen HMI in Siemens TIA Portal to automate control of six production parameters in a bottling plant mixing process, reducing manual operator intervention by $\sim 40\%$ and cutting fault detection response time by $\sim 30\%$.
- Benchmarked deep learning (LSTM) and machine learning (Random Forest) models on the NASA C-MAPSS dataset for time-series predictive maintenance; performed feature engineering (12 features) for remaining useful life (RUL) prediction, achieving 92% accuracy with LSTM vs 87% with Random Forest, and identifying trade-offs between predictive performance and deployment speed ($\sim 8\times$ faster inference).

INTRAINZ INNOVATION PVT. LTD. May 2023 – Jul 2023
Robotics Intern

- Designed and tested a 3-DOF robotic claw mechanism, achieving ± 0.5 mm positional accuracy through PID control and multi-sensor fusion.
- Documented dynamic model, kinematic equations, and PID tuning methodology in a structured technical report; findings used to guide iterative design improvements across 3 prototype iterations.

Skills

CAD/CAE Tools: SolidWorks, CATIA, Creo, AutoCAD, Fusion 360, Ansys, OpenFOAM, Siemens NX
Engineering Domains: Finite Element Analysis (FEA), Computational Fluid Dynamics (CFD), Geometric Dimensioning & Tolerancing (GD& T), Hydraulic & Pneumatic Systems, Kinematics & Dynamics
Programming Skills: MATLAB, Python, C, C++
Other Tools: MS Office, Canva

Projects

PINN-Based Study of Jet Impingement on an Isothermal Cylinder 

- Designed multi-objective physics-based loss functions and optimized neural network training using Adam and L-BFGS optimizers; generated CFD-style velocity, pressure, and temperature field visualizations with Nusselt number analysis, achieving $< 2\%$ error compared to benchmark CFD results.
- Implemented a 4-hidden-layer fully connected PINN ($2 \rightarrow 128 \rightarrow 128 \rightarrow 128 \rightarrow 128 \rightarrow 4$) with Xavier initialization and tanh activations in PyTorch; trained on 7,000 collocation points and 1,200 boundary points across inlet, outlet, cylinder wall,

and channel boundaries, enforcing no-slip and isothermal wall conditions ($T_{wall} = 0$, $T_{jet} = 1$) via weighted boundary condition loss ($\lambda = 300$).

Jet Impingement Cooling of a Heated Cylinder — CFD Validation (Ansys Fluent)

- Replicated a published numerical study on jet impingement cooling of a heated cylinder in crossflow ($Re = 20,000$) using a 2D axisymmetric $k-\varepsilon$ turbulence model in Ansys Fluent with SIMPLE pressure-velocity coupling and a structured, mesh-independence-verified grid.
- Validated local and stagnation Nusselt number distributions against published experimental data, confirming reliable CFD setup and boundary condition methodology consistent with the benchmark study.

Von Kármán Vortex Shedding over a Circular Cylinder (Ansys Fluent)

- Simulated unsteady laminar flow over a 2D circular cylinder at $Re = 200$ using a transient solver in Ansys Fluent with a structured, near-wall-refined mesh and time-dependent boundary conditions to resolve the periodic shedding cycle.
- Captured periodic Von Kármán vortex shedding and wake instability through vorticity contour visualization, consistent with classical bluff-body wake behavior in the laminar periodic shedding regime.

Certifications

- Supervised Machine Learning - Stanford Online
- Bajaj Manufacturing System CP - Bajaj Auto Ltd.
- Bajaj Manufacturing System ACP - Bajaj Auto Ltd.

Achievements

- **"Physics-Informed Neural Networks for Predicting Temperature Distribution in Engine Cooling Fins"** — Accepted at ASSET 2026 (presentation pending), co-authored with Dr. Nikhil Sohoni, Bajaj Institute of Technology.
- Smart India Hackathon 2023 Runner-Up — Government of India national competition; secured a top 5 finish among 390 student teams across India.
- Shiksha Mandal Badminton Champion — 3× consecutive first-place winner; Interzonal Runner-Up.